



Damage Detection Using Local Dominant Frequency of Pulse-Echo Laser Ultrasonic Waves

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Abstract. This paper presents the Local Resonance Frequency Map (LRFM) algorithm that provides an instant interpretation of structural features and defects using an estimation of local dominant frequency of the pulse-echo laser ultrasonic waves at a fine grid of spatial sampling points. A Q-switched laser and a laser Doppler vibrometer are used for generation and sensing of laser ultrasonic waves, and a two-axis translation stage is used for raster scanning of the combined generation and sensing laser beams. Experiments show that LRFM works well for damage detection in various structures from metallic to composites specimens.

Keywords: Local resonance frequency map · Pulse-echo laser ultrasonic waves

1 Introduction

Laser ultrasonic techniques for Non-Destructive Testing (NDT) of aeronautic industry have been developed for several decades. Initial studies in the 70s and 80s [1–3] showed strong potential for industrial applications. The pulse-echo mode of ultrasonic waves is showed to be the most effective method for NDT since it requires only access to one surface of the structure, in contrast to the through-transmission mode where two sides of access is needed; although the latter is more effective for detection of adhesively disbond [4]. In the 90s, several groups investigated the use of laser NDT for composite structures in cooperation with organizations such as Lockheed Martin and US Air Force [4–7]. Today, there are several companies provide commercial laser NDT: PaR Systems [8], Tecnar Automation [9], iPhoton Solutions [10], and Tecnatom [11]. In academic, this topic is also attracted much interested with several laboratory scale systems [12–14].

The key idea of the mentioned laser NDT systems [8–14] is to combine an extreme short pulse laser excitation with a sensing laser beam; and use a scanning mechanism to scan the combined beam: such as 5-DOF-axis robot [12], mirror scanner [8–11], and mechanical scanner XY stage [13, 14]. Each scanning mechanism has its own advantages as well as limitations. The use of a 5-DOF-axis robot for scanning allows inspection of very complex shaped objects such as small radius of curvature structures; however, the scan speed is slow with the max scan speed in [13] is 30 Hz. Mirror

scanner provides extremely fast scan speed (up to kHz range), also it is very compact. On the other hand, one issue of mirror scanner is that when the scan area is relatively large, the angle of incidence of the sensing beam may significant large near the edge of the scan area, which causes the signal quality is not uniform across the scan structure. In the case the structure is flat or mildly contoured, the mechanical scanner XY stage avoids the issue with high angle scan though it is much bulkier compared to the mirror scanner [14].

In this paper, a novel algorithm is presented for imaging damages of structure so-called Local Resonance Frequency Map (LRFM). It maps the dominant frequency of the pulse-echo laser ultrasonic waves at each point in the scan area. The result is a 2D image instead of 3D movie format in a C-scan movie. This algorithm provides an instant interpretation of structural features and damages since no input is necessary for LRFM, meaning that the result is shown immediately after finish scanning without any further input from user. Also LRFM has physical meaning instead of an arbitrary color-scale in a C-scan movie.

This paper is organized as follows. The next section presents the Local Resonance Frequency Map algorithm. Next, we apply this algorithm to detect damages of metallic and composite specimens. The final section is the conclusions.

2 Local Resonance Frequency Map Algorithm

When the laser focusses on the material, the laser irradiation generates bulk waves penetrated through the thickness h of the structure. The bulk waves reflected back and forth between two surfaces has the dominant frequency $\hat{f} = h/2v$ where v is the longitudinal velocity of the ultrasonic waves [14]. Local Resonance Frequency Map (LRFM) algorithm maps the dominant frequency of the pulse-echo laser ultrasonic waves at each point in the scan area. The flowchart of the algorithm is shown in Fig. 1. First, the time-space domain data $u(x,y,t)$ is transformed to frequency-space domain data $U(x,y,f)$ using 1D- Fourier transform. Next, for each frequency-domain signal from each scanning point, the maximum frequency is calculated. Finally, the 2D image containing all maximum frequency of the scan area is plotted.

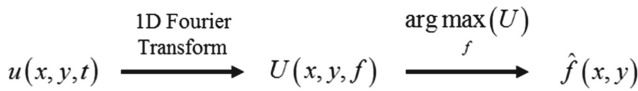


Fig. 1. Flowchart of Local Resonance Frequency Map (LRFM) algorithm

3 Experiments and Results

3.1 Aluminum Specimen with Thinning Area

The Pulse-Echo Ultrasonic Propagation Imager (PE-UPI) system [14] was used for inspection of a 4-mm aluminum plate with thinning area (100 mm $\times$ 200 mm) at 50%

thickness reduction. Picture of the specimen is shown in Fig. 2(a). The scan process were performed on the opposite side of the wall-thinning. The parameters of the experiment were set as follows. The laser was set to 5.85 mJ per pulse and 500 Hz pulse repetition rate. The standoff distance from the scanning laser heads to the specimen was 840 mm. The Q-switched laser beam impingement on the specimen had a diameter of 2.23 mm. The scanning interval of the translation stage was set to 0.52 mm on both the horizontal and vertical axes. The sensitivity of the LDV was set to 50 mm/s/V.

Figure 2(b)(c) show C-scan results at different time points. The damage area can be clearly visualized; however, there is no conclusion about the thickness of the damage area compared to the intact area. Figure 2(d) shows the LRFM result. The LRFM shows that in the thinning area, the dominant frequency is approximately 1.5 MHz and in the intact area, the dominant frequency is approximately 0.75 MHz. This result is showed to be perfectly agreed with the theory as presented in Sect. 2 where the thinning area has 50% thickness reduction.

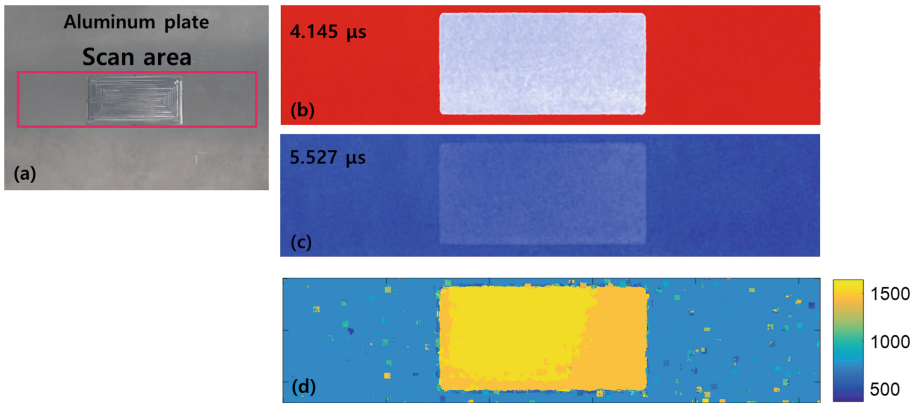


Fig. 2. (a) Aluminum plate of 4-mm thickness with 50% thickness reduction at thinning area, (b) (c) C-scan results captured at 4.145 μ s and 5.527 μ s, (d) Local Resonant Frequency Map, the unit of color bar at the right side is kHz

3.2 Composite Specimen

The PE-UPI system was used for inspection of a composite specimen with ply defects made during its manufacturing. Picture of the specimen is shown in Fig. 3(a). The parameters used for testing this specimen was identical with the previous test with the aluminum specimen, except the laser energy is set at lower level 2.5 mJ per pulse to avoid surface damage of the composite specimen.

Figure 3(b)(c) show the C-scan results at different time points. The C-scan image at 10.700 μ s only shows one ply defect at the top-right corner. The C-scan image at 78.400 μ s shows all three ply defects; however the image of the defects is not clear since the ultrasonic waves is highly diminished after multiple back-and-fourth

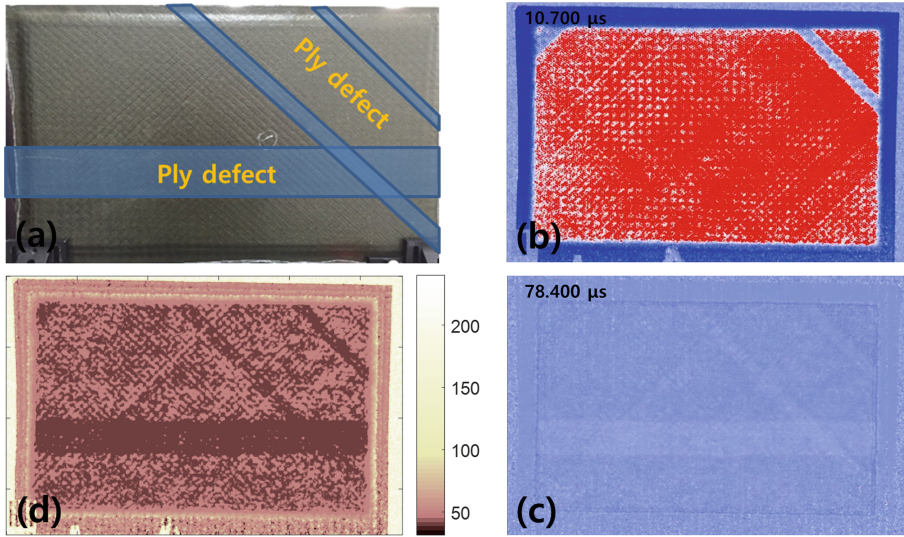


Fig. 3. (a) Composite specimen with ply defects, (b) (c) C-scan results captured at 10.700 μ s and 78.400 μ s, (d) Local Resonant Frequency Map, the unit of color bar at the right side is kHz

reflections. Figure 3(d) shows the LRFM result, which captured all ply damage in a single frame with much better resolution of the defects.

4 Conclusions

The novel algorithm developed in this paper provides an instant interpretation of structural features and defects. No additional input is necessary for LRFM, meaning that the result is shown immediately after scanning. It gives the users of the PE-UIP system the preliminary result before proceed to use other algorithms such as VTWAM [15], or UWI [16] for deeper analysis. Another advantage of the LRFM algorithm is that it has physical meaning instead of arbitrary color-scale in a C-scan movie, which allows possible evaluation of structural thickness. Experiments show that LRFM works well in various structures from metallic to composite specimens.

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